APPROXIMATING OBJECTIVE AESTHETIC ASSESSMENT WITH ARTIFICIAL NEURAL NETWORK

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ABSTRACT

This article discusses the possibility of using an Artificial Neural Network to provide aesthetic assessments of automatically generated counterpoints. It presents in detail several different architectures of such networks and analyses training strategies. In particular, it demonstrates how to train such a network for providing objective assessments (such as those prescribed by course materials on counterpoint) and why this approach is not entirely practical. It also observes that Artificial Neural Networks may, in fact, exhibit some form of machine subjectivity.

INTRODUCTION

Human perception of beauty is subjective. Each individual perceives the same piece of art differently. Moreover, the same individual can provide different aesthetic assessments of exactly the same piece of art at different times. Human impressions are biased by various criteria, not necessarily related to art. Culture, time of the day, mood, politics, religion, relationship with the author, all this can affect the perception of beauty.

Nevertheless, it is possible for a number of humans to agree on some universally accepted attributes of a piece of art responsible for beauty. For example, the rules of counterpoint can objectively determine if a musical composition (i.e. a counterpoint) is *beautiful* or *ugly*. In particular, the number of repetitions or the number of parallel consonances can be used as a numerically expressed aesthetic score.

The objective aesthetic assessment, based on clearly defined rules, can be used as a basis of a computer program. Such an artificial judge is then able to assess pieces of art automatically. Typically, it will provide exactly the same judgements of the same piece of art every time without any bias. It may also assess several different art pieces as equally beautiful, while human observers will value them differently. In other words, when computer says that these two melodies are equally beautiful, I may say that I like this one better than the other, but I don't know why I like it better. Perhaps it resembles something I have heard before, which I liked a lot?

A computer program is usually nothing more than a series of clearly defined steps, a prescription how to achieve something, solve an equation and in general, is usually ultra objective. From the programmer's perspective, computer programs always do exactly what they were instructed to do, mercilessly. Within this context, is it possible to create an artificial aesthetic judge, which will have some subjectivity? In particular, is it possible to train a computer program in aesthetics to the extent that we, as humans, will explain why two pieces of art are equally beautiful, but the program will still say that it likes one of them better then the other? Is it possible to create a computer program, which will calculate a different aesthetic assessment of the same melody at different times and will be biased by its previous experiences?

The human brain is totally different to an electronic computer. While we don't know exactly how the brain works, we know it is composed of neurons. The hypothesis is that an artificial neural network may in fact exhibit some form of machine subjectivity.

Neural networks described in this paper were trained to reproduce the objective aesthetic assessment of automatically generated counterpoints. It has been observed that during the experiments neural networks exhibited some level of machine subjectivity, which are further analysed and explained.

BACKGROUND INFORMATION

Artificial Neural Networks constitute well established technology. They are particularly good for detection and classification of patterns, picture recognition (such as handwriting), signal processing and filtering [8]. They have already been used in the field of music and music research [5]. In particular they were used for automatic generation of musical compositions [2, 3, 4] and recognising musical genre [6].

Artificial Neural Networks were also used as Fitness Functions in conjunction with Genetic Algorithms [2, 3]. However, this approach has proven to be problematic [1].

AESTHETIC ASSESSMENTS

For the purpose of the experiments described in this paper, aesthetic assessments were clearly divided into two distinctively different categories: objective and subjective.

Objective Aesthetic Assessment

In the context of this paper, the objective assessment can be defined as a numeric score (or a combination of scores), which is universally accepted and agreed upon by certain group of people, such as composers or teachers. It is also independent of individual preferences of each human. For example, the number of parallel imperfect consonances in a counterpoint or number of repeated notes can be considered as an objective assessment (or part of it). Such precisely defined criteria can be used to described two different counterpoints as equally beautiful, because they contain exactly the same number of repetitions and parallel imperfect consonances.

The aesthetic guidelines were extracted from the course material for the undergraduate counterpoint classes at the University of Adelaide, that are taken by composers [7].

Subjective Aesthetic Assessment

This objective aesthetic assessment differs substantially from a subjective impression of each human listener. The main reason for it is that the musical notation is merely a representation of a piece of music rather than the music itself. Hence, we do not assess the music notation, but the music that the notation describes or represents. Moreover, each individual human listener will most likely assess music differently than his fellow listeners. The reasons may also include personal bias toward or against the creator of the composition, current mood and whatever the listener did or experienced recently. In other words, if there are two counterpoints with exactly identical objective aesthetic scores, one of them may be perceived as more pleasing than the other. And we, as humans, will have difficulty to formulate a believable justification for our preference, apart from saying that it just "sounds better".

Assessment Calculation

The rules of objective aesthetic assessment can be easily turned into a computer program (as long as they are clearly described). Such program will be able to identify counterpoints, which are identically pleasing. We may be able to say that we like them differently. However, we may not be able to provide any justification as to why we like one better than the other. This kind of subjective assessment will also be beyond machine comprehension.

SUBJECT OF THE EXPERIMENT

All the counterpoints used in the experiments were automatically generated by an expert system written in Prolog [9]. In particular, the system calculated all the valid counterpoints for a given cantus firmus. The number of counterpoints was frequently very high (typically exceeding one million for a 12-note cantus firmus). To decrease the computational requirements and speed up the experiments, the number of all valid counterpoints was further reduced by limiting the cantus firmus to 4 notes only and locking in one particular rhythm. This way it was possible to reduce the number of all valid counterpoints to less than 100.



Figure 1. Cantus Firmus used in the experiments.

Each of the valid counterpoints was assigned an objective score consisting of 3 values: the highest number of repeated notes, number of parallel imperfect consonances and the total number of parallel/contrary motions (where each oblique movement was scored at 1 and each parallel movement was scored at 2). Values closer to ZERO were considered *nice*, while higher values represented increasing levels of *ugliness*. These counterpoint attributes were selected based on the coursework material used at the University of Adelaide for teaching counterpoint.

The aforementioned scores were calculated automatically by the program. From the machine point of view, the selection of counterpoint attributes used for calculating the objective aesthetic score is arbitrary. As long as each counterpoint can be assigned a value calculated in any way, it is good enough for the purpose of the experiment.

INITIAL EXPERIMENTS

The purpose of the first set of experiments was to connect counterpoints to the input of a neural network to see how well it could learn to approximate objective aesthetic scores. Each counterpoint together with its cantus firmus, was encoded numerically to represent note pitches ranging from 0 to 128. Rhythmical values were broken into a series of crotchets played legato. Hence, a semi-breve became 4 crotchets with the same pitch value.

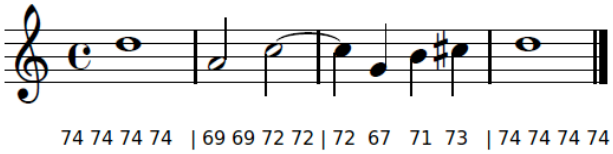


Figure 2. Counterpoint and its numerical representation of pitch.

Training a Single-Neuron Network

The first experiment involved a neural network consisting of a single neuron. It was trained to recognise the number of repeated notes in the counterpoint.

After a series of training sessions it was observed that a single neuron is capable of approximating the accurate detection of the number of repeated notes to some extent. In particular, while the number of repeated notes is always an integer number, the neuron was reporting values with fractional parts. However, those numbers were close enough to the expected value most of the time, although there were several cases of mis-approximations.

It was possible to focus the training routine on the mis-approximated cases and correct them. However, this was done at the expense of correct detections becoming less accurate. In general, it seemed impossible to train the neuron to provide completely accurate results all the time.

Training a Three-Neuron Network

The second experiment tried to replicate the entire objective assessment, which consisted of 3 parts (including repeated notes, parallel consonances and parallel motions). As the output of the neural network was expected to have 3 numbers, it needed to consist of 3 neurons.

After running a training procedure, the results were identical to the experiment involving a single neuron. In particular, focusing the training on increasing the accuracy of one particular counterpoint was done at the expense of other counterpoints.

Training a Multi-Layer Network

The third experiment involved a multi-layer architecture of a neural network. In particular a 3 layer network was tested. It contained 32 neurons in its first layer (input), 16 inside the hidden layer, and 3 neurons for the output.

During the experiments it was observed that the results were similar to those achieved by single-layer neural networks. The accuracy of detections became slightly higher. However, the training time increased dramatically.

Outcomes

All experiments demonstrated that a artificial neural network with simple architecture is hardly capable of approximating an objective aesthetic assessment accurately. Although it frequently generates results that can be considered as *close enough*, there are still cases that are too inaccurate.

When considering this result, the situation may be interpreted as a similar mechanism to a subjective aesthetic assessment. It could be perceived as the neural network exhibiting some form of subjectivity. Primarily, the results coming out from the neural network are always fractional, while the original objective values are integers. This could be described as the neural network liking something *a little bit more* or *just a bit less* than the rules suggest. Secondly, in some cases the neural network assessment differs drastically from what is expected. This can be characterised as *a strongly subjective impression* of the neural network.

Although the experiments show that the neural network demonstrates inaccuracies, there is still the possibility that 100% accurate objective response may be achieved if enough training is provided. In such a case, the neural network would loose its *subjectivity*. The next experiments focus on this situation.

ACHIEVING 100% OBJECTIVITY

This set of experiments focuses on achieving completely accurate reproduction of the objective aesthetic assessment.

Counter-Propagation Neural Network

The architecture selected for this experiment is called a *Counter-Propagation* neural network. It consists of two layers of neurons: the Kohonen layer and the Grossberg layer. The Kohonen layer has slightly modified functionality to the typical neural network. In particular, from all its neurons, the one with the highest output signal is selected as a *winner*. The value of this winning neuron is then changed to 1.0, while the responses of all the other neurons are set to 0.0. The purpose of this layer is to recognise and classify input patterns. As the training set consisted of 49 counterpoints, this layer must have at least 49 neurons. Each of them is trained to recognise one particular counterpoint. The purpose of the second layer (Grossberg) is to reconstruct the expected output signal, which is the objective aesthetic assessment itself.

Training of the Kohonen layer

The training of the first layer of *Counter-Propagation* neural network is still a tedious process. In the end, the Kohonen layer is fully capable of recognising counterpoints with 100% accuracy. However, the fact that it is capable of doing so does not guarantee success every time. A mathematical proof is a necessary ingredient here.

Taking advantage of the fact that the input to the Kohonen layer is always normalised, the content of each neuron can be mathematically calculated rather than trained. In particular, all the counterpoints from the training set can be normalised and used to populate the neurons without any further modification. Hence, it proves that it is possible to train such a layer to be 100% accurate.

Training of the Grossberg layer

The training of the second layer of *Counter-Propagation* neural network is a trivial task. As the input to the Grossberg layer will always consist of 1.0 generated by a singled neuron (and 0.0 on all others), the number of training repetition necessary is small. It can also be mathematically calculated rather than trained.

Outcomes

It has been observed that the Counter Propagation neural network is capable of accurate reproduction of the objective aesthetic assessment. However, The Kohonen layer must contain the number of neurons equal (or greater) to the number of all possible counterpoints. This number can be very high. A 12-bar melody, consisting of 48 crotchets, where each can be one of 12 pitches (assuming we put a limit of one octave), with completed disregard of the rules of counterpoint, can yield 6,319,748,715,279,270,675,921,934,218,987,893,281,199,411,530,039,296 results. This is much more than the total number of neurons present in a human brain, which is estimated at 100,000,000,000. In fact, it exceed the number of neurons in the entire human population on Earth (7 billion) by several orders of magnitude.

EXPLORING SUBJECTIVITY

The following set of experiments explores the *subjective* behaviour of the neural network's output. In particular, it focuses on creating a situation where the difference between expected and real output changes its sign. In other words, it may be interpreted that the neural network *changed its mind* and now it likes one particular counterpoint *a bit less*, while it used to *like it a bit more* than objectively prescribed.

The experiments were conducted on a two-layer neural network. It contained 16 neurons in its input layer. This number was arbitrarily selected. The number of neurons in the output layer was dictated by the size of the objective assessment vector, which is 3.

Three different training sets of counterpoints were created. All 3 sets consisted of florid counterpoints locked to one particular rhythm. The total number of counterpoints in each set was: 12, 49 and 84.



Figure 3. One counterpoint from the first set.



Figure 4. One counterpoint from the second set.



Figure 5. One counterpoint from the third set.

Experimenting with Different Sets of Counterpoints

In this experiment three identical neural networks were trained to approximate objective aesthetic assessments from each separate set. After relatively good detections were achieved, the networks swapped the training sets and it was observed that the results were too random and the number of mis-assessments was almost 100%.

Experimenting with a Pre-Conditioned Neural Network

In this experiment an already trained neural network was re-trained for detecting different counterpoint sets. However, after the training the network seemed completely to forget its prior training. The results were no different to training routines starting with random state neural network.

Changing the Order of Counterpoints within a Training Set

This experiment also was unsuccessful. After the training routing was complete, the results generated by the neural network were identical with those obtained with unchanged order.

Experimenting with Continuous Learning

In this experiment the neural network was continuously learning, while generating results at the same time. It became evident that prior expositions of counterpoints influenced the neural network's output if the learning factor was high enough. In the following example, the learning factor was set to 0.2. This result was obtained from the training set consisting of 12 counterpoints. The change of sign of the neural network's error became evident when the order of counterpoints within the training set was changed. The table below shows only the output from the second neuron, which represents the number of repeated notes within the counterpoint. This number was normalised to the range from 0.0 to 1.0, hence the fractional values of 0.625 and 0.75. The other components were omitted for clarity.

ID	Expected	Returned	Difference
0	0.625	0.634	0.009
1	0.625	0.631	0.006
2	0.625	0.631	0.006
3	0.625	0.623	-0.002
4	0.625	0.617	-0.008
5	0.625	0.616	-0.009
6	0.625	0.622	-0.003
7	0.625	0.625	0.000
8	0.625	0.623	-0.003
9	0.625	0.625	0.000
10	0.750	0.747	-0.003
11	0.750	0.748	-0.002

Table 1. The results of approximated number of repeated notes within counterpoint with original order of counterpoint.

ID	Expected	Returned	Difference
9	0.625	0.607	-0.018
6	0.625	0.633	0.008
5	0.625	0.620	-0.005
7	0.625	0.629	0.004
8	0.625	0.617	-0.008
10	0.750	0.747	-0.003
11	0.750	0.748	-0.002
4	0.625	0.628	0.003
0	0.625	0.628	0.003
1	0.625	0.629	0.004
2	0.625	0.633	0.008
3	0.625	0.623	-0.002

Table 2. The results obtained from the randomly re-ordered training set.

Interpretation of the Outcomes

In the above experiment, the difference between neural network output and the expected objective aesthetic assessment (i.e. the normalised number of repeated notes) has changed its sign in two cases. This result can be interpreted as the neural network *changed its mind* about *liking* or *disliking* a counterpoint.

According to the contrapuntal recommendation, it is desired to have a variety of different notes in the counterpoint. Therefore the number of repetitions of notes should be as small as possible. In the context of the above experiment, the lower the value of the normalised number of repeated notes, the better the aesthetic result. Hence, the positive difference represents *liking a bit less*, while the negative difference indicates *liking a bit more*. In particular, in the example above, the neural network changed its *liking* of counterpoints no. 4 and 6, after changing the order within the training set.

CONCLUSIONS

It is mathematically possible to train (or populate) a neural network to be able to provide accurate objective assessments. However, the size of such a network is too big for any practical use by many orders of magnitude.

In a typical implementation, where number of neurons and neural layers is limited to manageable amount, the neural network is able to approximate objective assessment with some small error. While this error is usually small enough, it can occasionally become too big and lead to mis-assessments. Also, the error is inversely proportional to the number of neurons.

During the experiments it also became evident that a neural network is erratic when assessing counterpoints which were not present in the original training set. This behaviour makes it difficult to user neural networks as meaningful fitness functions for evolutionary generative music [1].

However, artificial neural networks may be useful in developing some form of machine subjectivity. Their output is always *a little higher* or *lower*, which is the result of some computational error. In a typical scenario, this error becomes lower during the training process. It is nevertheless possible to keep this error continuously evolving by careful manipulation of the learning factor and maintaining the training process indefinitely. Moreover, it is possible to change the sign of this error by changing the order of expositions within the training set.

The aforementioned behaviour is to some extent similar to human subjective perception of beauty and ugliness. Our brain continuously learns. Hence, our aesthetic preferences evolve. Also, our previous experiences, even changes to our daily routine influence our judgements. While we are influenced by our human activities (such as food, entertainment, social problems, ideas, etc.), computer programs can be influenced by their previous computer activities. In this case it can be the order of counterpoint expositions.

FURTHER RESEARCH

The possible direction for further research may focus on testing more complex neural network architectures, such as those with feedback loops, including Hopfield's networks and Boltzmann machines [8]. Very large training sets could be examined as well. In particular, sets consisting of millions of possible counterpoints. It may also be interesting to investigate if it is possible to use neural networks to develop some form of machine subjectivity within bigger systems involving genetic or evolutionary algorithms.

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